

0	5	.	1
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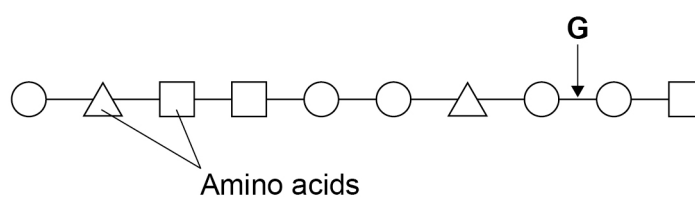
Describe how the structure of glycogen is related to its function.

[4 marks]

[illegible]

Figure 5 shows the primary structure of part of a polypeptide. Each shape represents an amino acid. Identical amino acids have the same shape.

Figure 5



0	5	.	2
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Name the type of peptidase which will hydrolyse the bond labelled **G** in **Figure 5**.

[1 mark]

0	5	.	3
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Give the number of different **R** groups in the polypeptide shown in **Figure 5**.

[1 mark]



A scientist used an enzyme to digest a polypeptide containing 101 amino acids. The digestion produced a range of smaller polypeptides.

The scientist determined the number of amino acids in each of the polypeptides produced. He also counted the number of polypeptides of each length.

Table 1 shows some of the scientist's results.

Table 1

Number of amino acids in polypeptide	Number of polypeptides of each length
5	2
6	
15	3
20	

0 5 . 4

Use the information in **Table 1** to calculate the number of polypeptides:

[2 marks]

6 amino acids in length _____

20 amino acids in length _____

8

Turn over for the next question

Turn over ►

